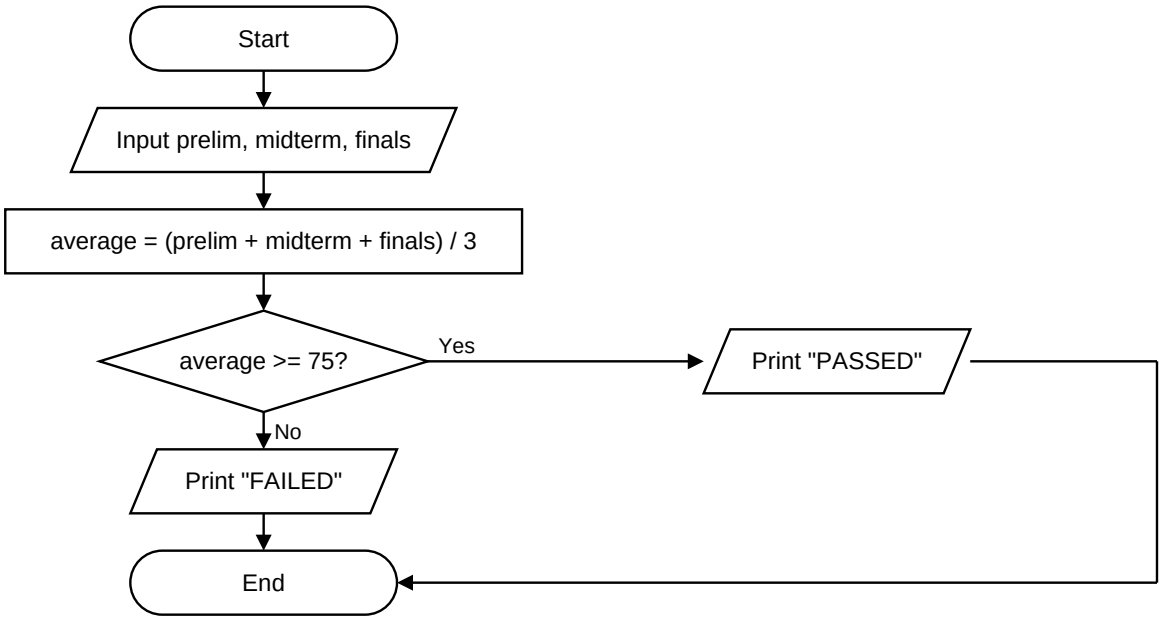
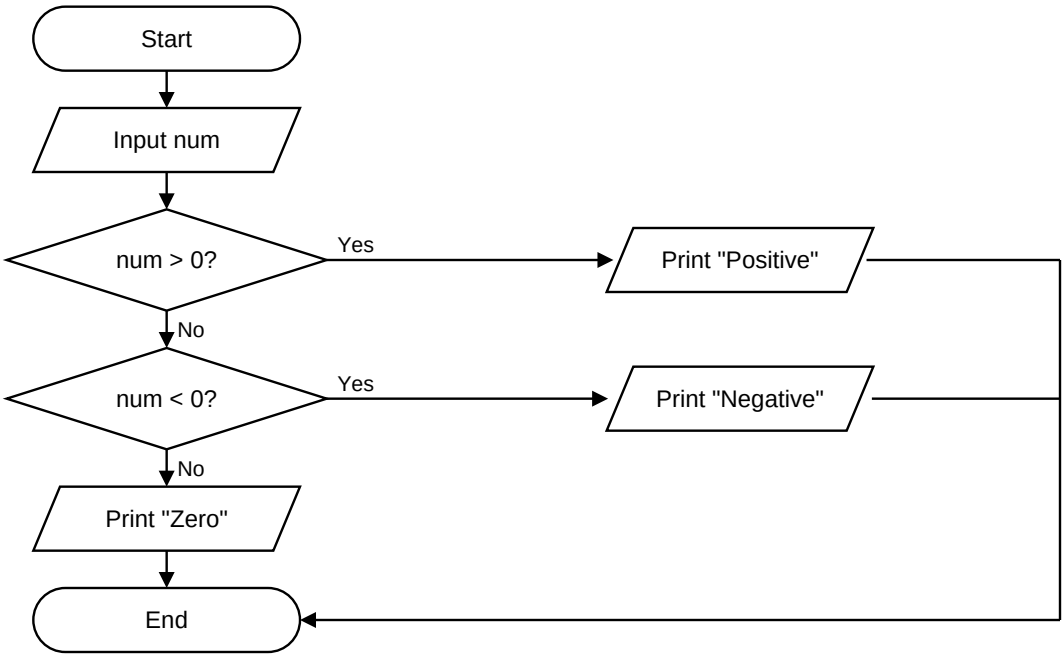
1. Revise your Grade flowchart in Sequence constructs by identifying the remarks whether the average grade is Passed or Failed. If average is greater than or equal to 75 then passed else failed.



Simulation table

Prelim	Midterm	Finals	Average	Average >= 75?	Remarks
95	91	93	93	Yes	Passed
70	72	74	72	No	Failed
75	75	75	75	Yes	Passed

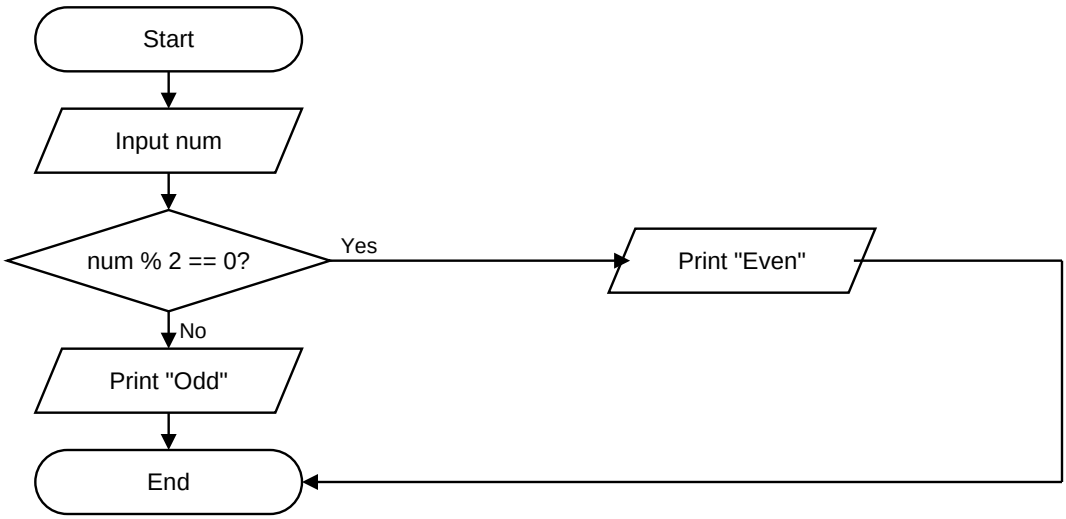
2. Create a flowchart that will input for a number then identify whether the number is positive or negative.



Simulation table

num	num > 0?	num < 0?	Remarks
30	Yes	-	Positive
-15	No	Yes	Negative
0	No	No	Zero

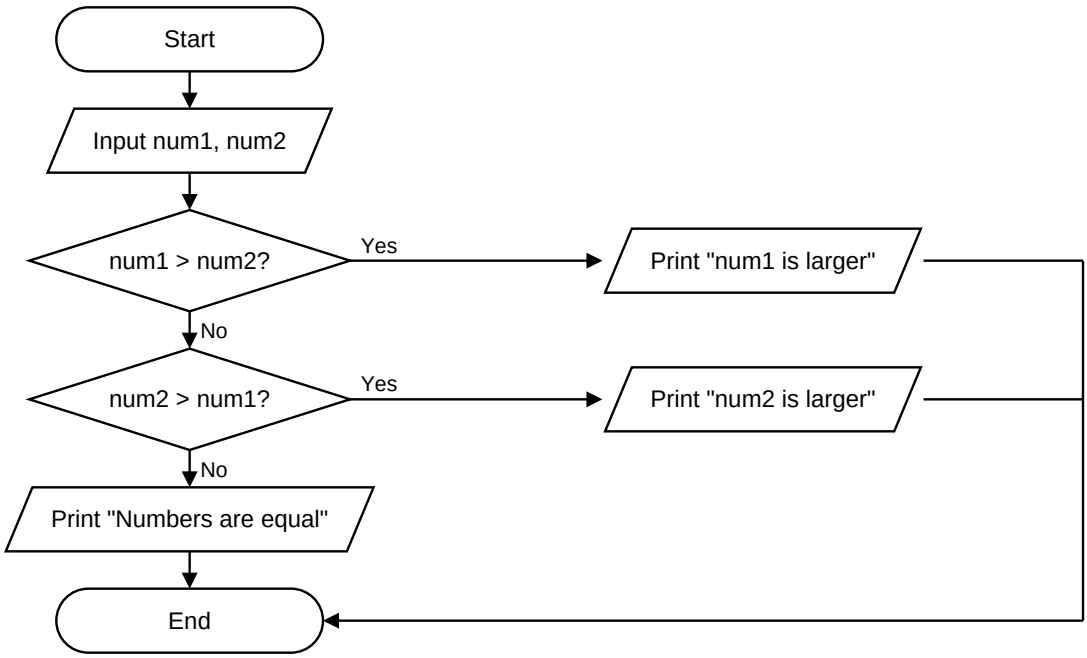
3. Create a flowchart that will input for a number then identify whether odd or even number.



Simulation table

num	num % 2	num % 2 == 0?	Output
30	0	Yes	Even
15	1	No	Odd

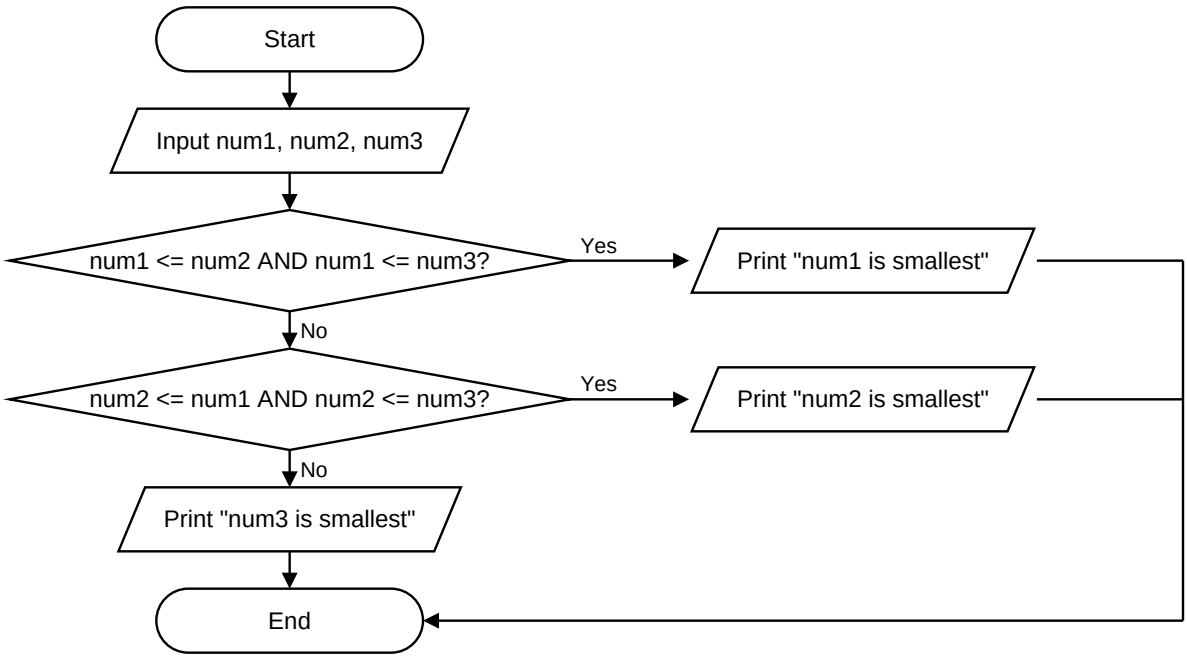
4. Create a flowchart that will input for 2 numbers then identify which of the two is the largest.



Simulation table

num1	num2	num1 > num2?	num2 > num1?	Output
30	15	Yes	-	30 is larger
18	28	No	Yes	28 is larger
20	20	No	No	Numbers are equal

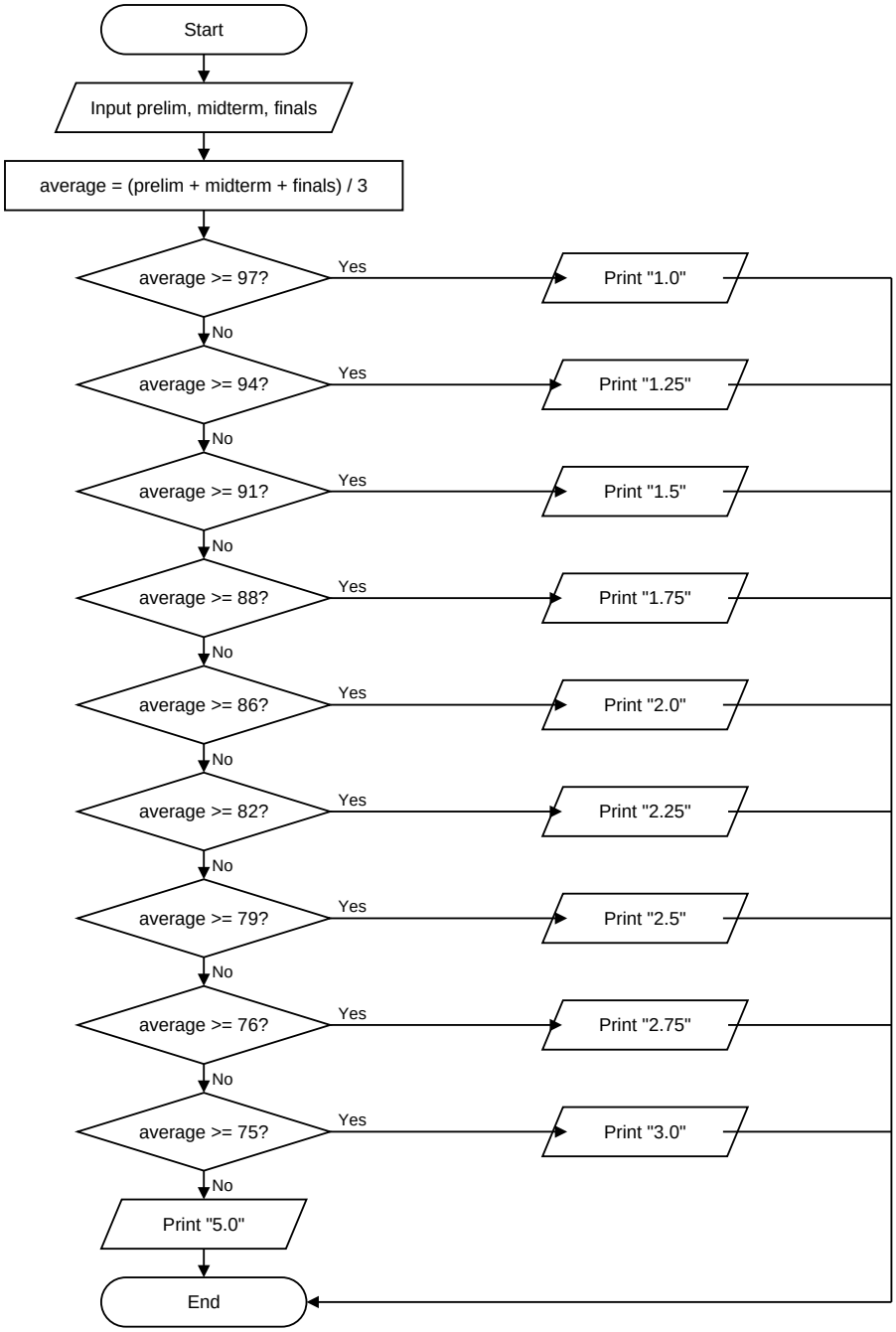
5. Create a flowchart that will input 3 numbers then identify the smallest.



Simulation table

num1	num2	num3	num1 smallest?	num2 smallest?	Output
7	10	14	Yes	-	7 is smallest
10	6	9	No	Yes	6 is smallest
12	9	4	No	No	4 is smallest

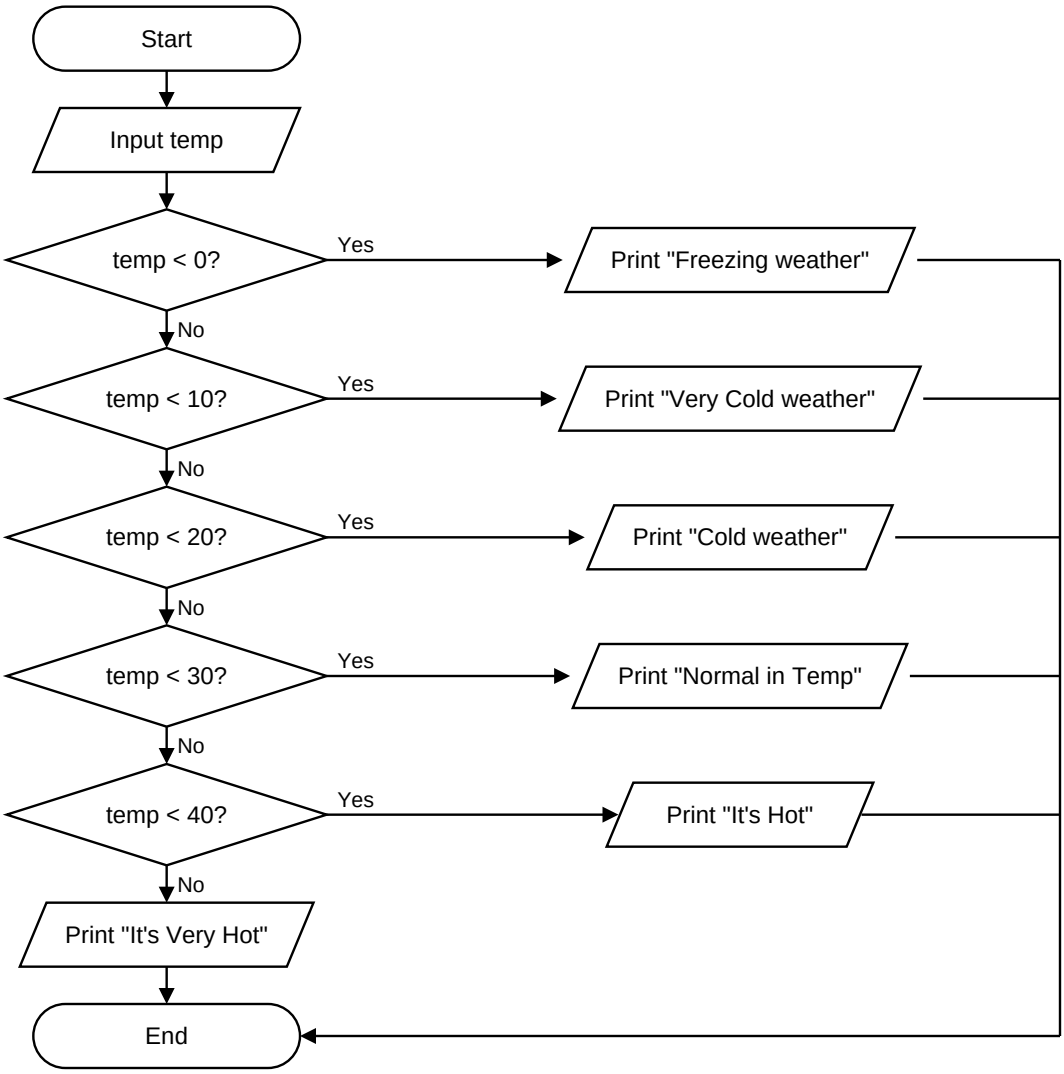
6. Revise exercise #1 in selection problems by determining the equivalent grade of the computed average: 97-100 = 1.0, 94-96 = 1.25, 91-93 = 1.5, 88-90 = 1.75, 86-88 = 2.0, 82-84 = 2.25, 79-81 = 2.5, 76-78 = 2.75, 75 = 3.0, else 5.0.



Simulation table

Prelim	Midterm	Finals	Average	First Yes	Output
98	93	94	95	>= 94	1.25
99	98	97	98	>= 97	1.0
80	78	79	79	>= 79	2.5
75	75	75	75	>= 75	3.0
70	72	74	72	none	5.0

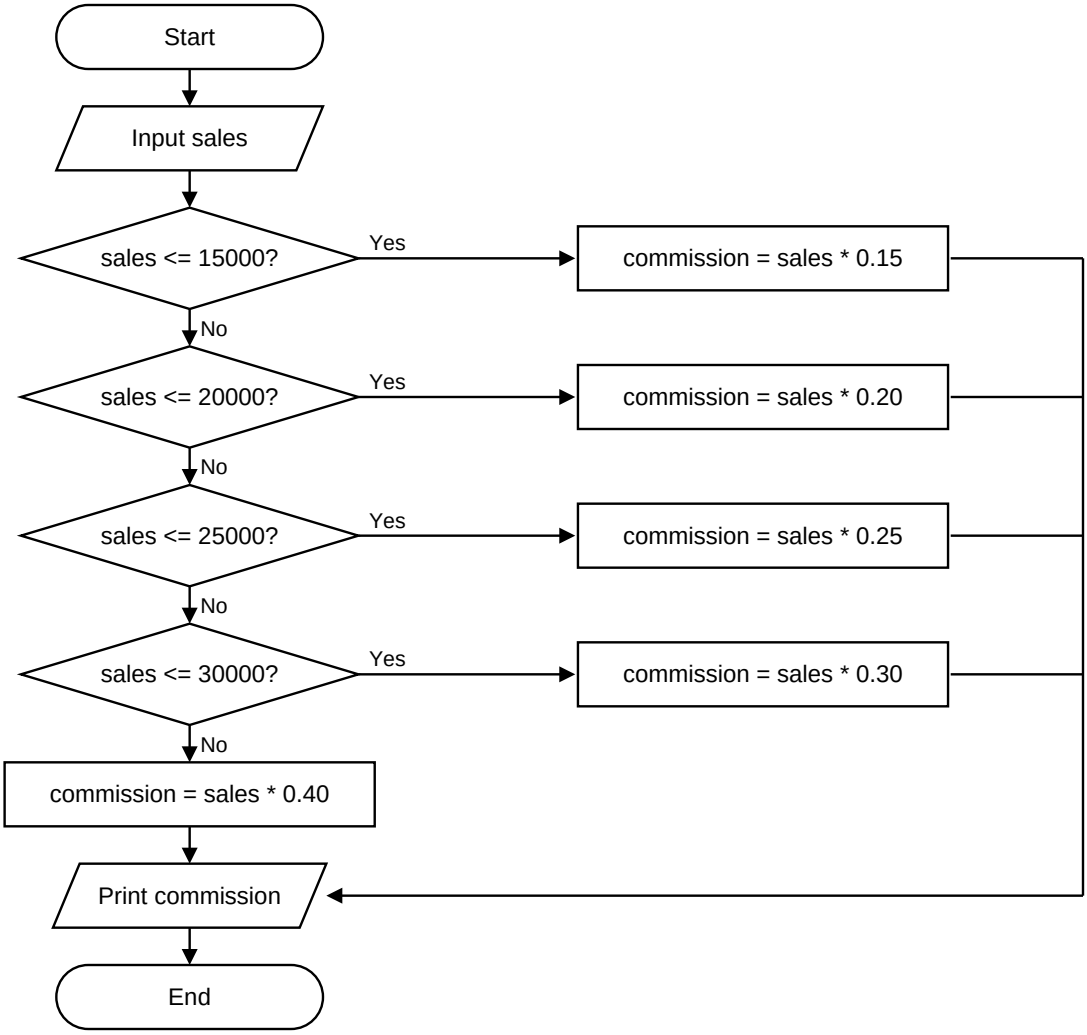
7. Create a flowchart to read temperature in centigrade and display a suitable message: Temp < 0 Freezing weather; 0-10 Very Cold; 10-20 Cold; 20-30 Normal; 30-40 It's Hot; >= 40 It's Very Hot.



Simulation table

temp	< 0	< 10	< 20	< 30	< 40	Output
-18	Yes	-	-	-	-	Freezing weather
8	No	Yes	-	-	-	Very Cold weather
18	No	No	Yes	-	-	Cold weather
28	No	No	No	Yes	-	Normal in Temp
38	No	No	No	No	Yes	It's Hot
45	No	No	No	No	No	It's Very Hot

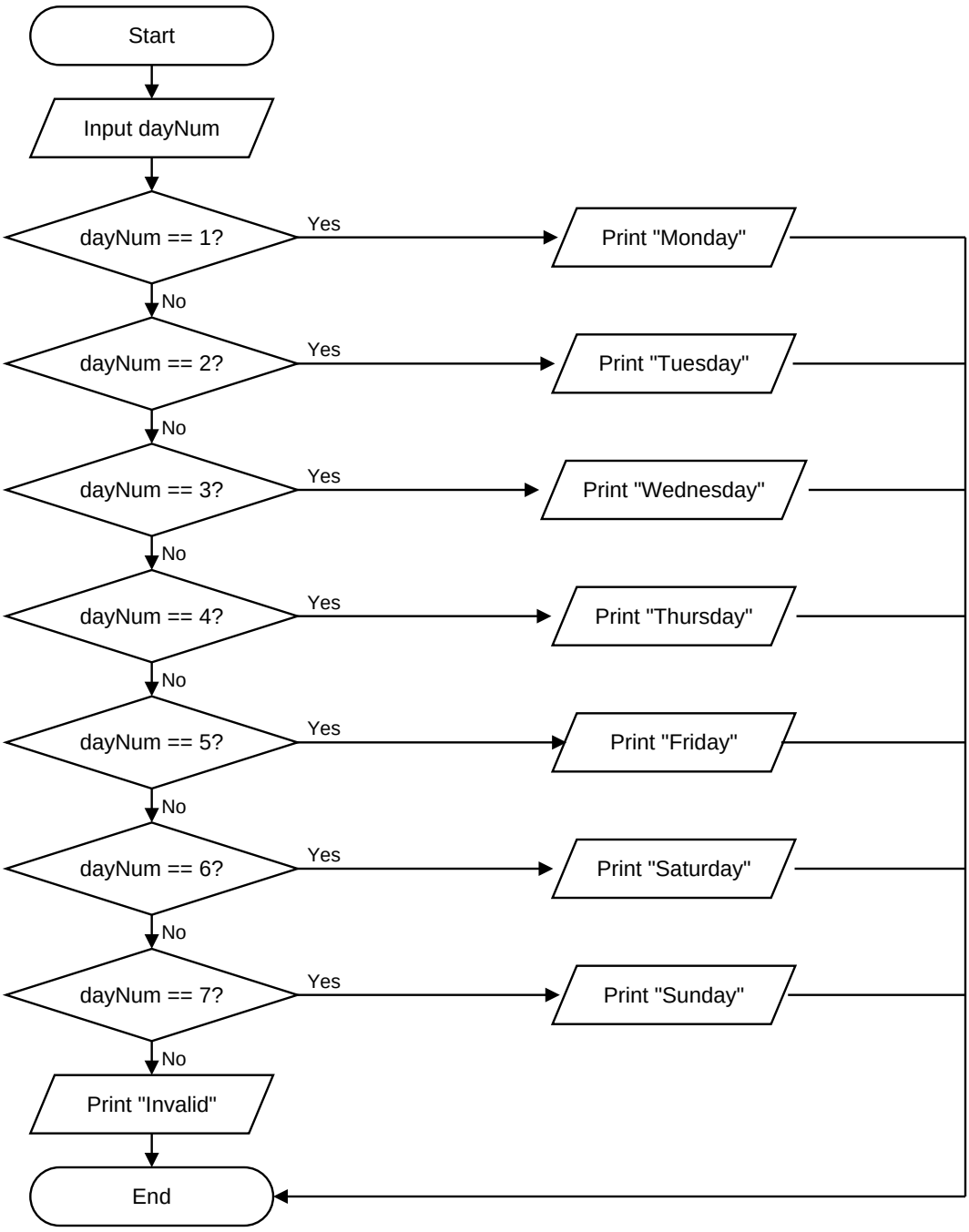
8. Revise your Sales problem in Sequence constructs by computing and displaying the sales commission based on: Sales <= 15,000 = 15%; <= 20,000 = 20%; <= 25,000 = 25%; <= 30,000 = 30%; > 30,000 = 40%.



Simulation table

sales	<= 15000?	<= 20000?	<= 25000?	<= 30000?	Rate	Commission
8000	Yes	-	-	-	0.15	1200
18000	No	Yes	-	-	0.20	3600
27000	No	No	No	Yes	0.30	8100
35000	No	No	No	No	0.40	14000

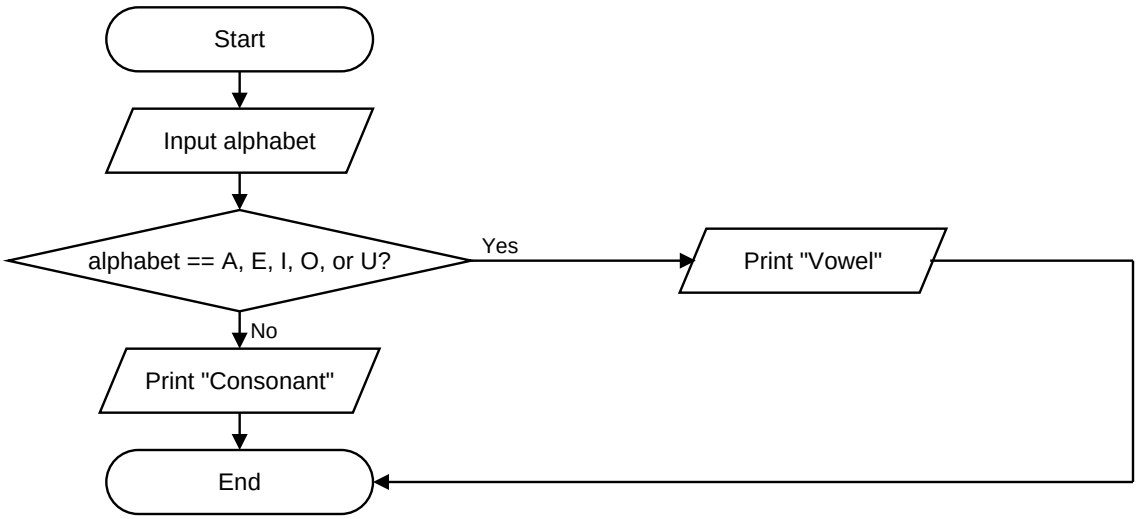
9. Create a flowchart to read any day number in integer and display the day name in word format.



Simulation table

dayNum	== 1	== 2	== 3	== 4	== 5	== 6	== 7	Output
7	No	No	No	No	No	No	Yes	Sunday
3	No	No	Yes	-	-	-	-	Wednesday
9	No	No	No	No	No	No	No	Invalid

10. Write a flowchart to check whether an alphabet is a vowel or a consonant.



Simulation table

alphabet	alphabet == A, E, I, O, U?	Output
R	No	Consonant
A	Yes	Vowel
U	Yes	Vowel